CASE STUDY FOR COMPUTATIONAL ENGINEERING

TRAFFIC FLOW BALANCE AT ROAD JUNCTIONS

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INTRODUCTION

Traffic congestion at **road intersections is a common challenge** in urban transportation systems.

At a four-way junction, vehicles enter from different directions and redistribute by turning left, right, going straight, or making a U-turn.

To analyze this behavior, traffic flow can be modeled using the conservation of vehicles, where total inflow equals total outflow.

In this study, a matrix formulation is used to represent vehicle redistribution at the junction.

To make the model more realistic, nonlinear congestion effects and concepts from the kinetic theory of traffic flow are incorporated. MATLAB is used to simulate the system and visualize the results through graphs.

TRAFFIC FLOW BALANCE

AIM/OBJECTIVES

- To represent traffic movement using a matrix formulation.
- To compute incoming and outgoing traffic flows at the intersection.
- To introduce nonlinear congestion effects to model road capacity limits.
- To study the relationship between density, velocity, and traffic flow using the kinetic theory of traffic.

PROBLEM DESCRIPTION

Consider a four-way road junction with traffic entering from:

- North
- South
- East
- West

Vehicles entering the junction can:

- Go straight
- Turn left
- Turn right
- Make a U-turn

The movement of vehicles satifies

Conservation Principle:

Total Inflow = Total Outflow

- A turning ratio matrix is used

MATRIX MODEL FOR TRAFFIC FLOW

MATRIX FORMULATION

Let the incoming traffic be represented as a vector:

$$Xin = \begin{bmatrix} N \\ S \\ E \\ W \end{bmatrix}$$

Where:

- N,S,E,W represent traffic entering from North, South, East, and West.

A **turning ratio matrix (T)** represents how vehicles redistribute to different directions.

Outgoing Traffic (Xout) = T x Xin

Sum of each coloum of T =1

CODE

solve_traffic_flows.m × Traffic_Flow_Final.m ×

/MATLAB Drive/COMPUTATIONAL ENGINEERING LAB/CASE STUDY/Traffic_Flow_Final.m

```matlab
clc; clear; close all;

% Rasesh J. Nair (23BME021) CASE STUDY ON "Traffic Flow Balance at Road
% Junctions

%% ------------------------------
% PART 1 : Linear Intersection Flow (LAB 3)    (please refer to the pdf for
% more clarity)
%% ------------------------------

% Incoming traffic vector (veh/min)
X_in = [250; 220; 200; 240];   % [North; South; East; West]

disp('Incoming Traffic (veh/min):')
disp(X_in)

% Turning ratio matrix T
% Row = Exit direction
% Column = Entry direction
% Order: [North; South; East; West]

```

COLUMNS (INCOMING ROAD)

ROWS (EXIT DIRECTION)

..._flows.m × Traffic_Flow_Final.m ×

.../COMPUTATIONAL ENGINEERING LAB/CASE STUDY/Traffic_Flow_Final.m

```matlab
T =[0.10  0.40  0.20  0.30;   % North exit
    0.40  0.10  0.30  0.20;   % South exit
    0.30  0.20  0.10  0.40;   % East exit
    0.20  0.30  0.40  0.10];  % West exit

% Each column sums to 1 (conservation check)
disp('Column sums of T (should be 1):')
disp(sum(T))

% Outgoing traffic (linear case)
X_out_linear = T * X_in;

disp('Outgoing Traffic without Congestion (veh/min):')
disp(X_out_linear)

```

For conservation of vehicles:

$$0.10 + 0.40 + 0.30 + 0.20 = 1$$

This means:

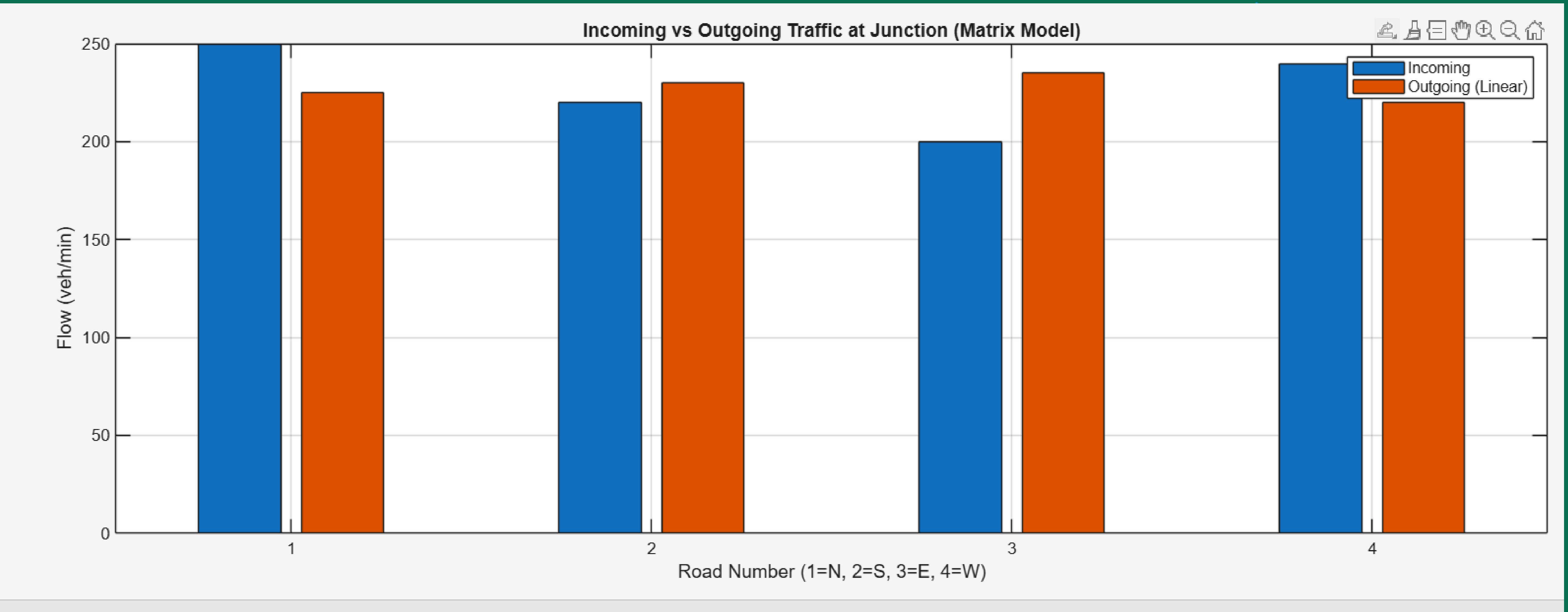
100% of vehicles entering must leave somewhere.

This ensures:

$$\text{Total Inflow} = \text{Total Outflow}$$

OUTPUT

```
figure

bar([X_in X_out_linear])

xlabel('Road Number (1=N, 2=S, 3=E, 4=W)')
ylabel('Flow (veh/min)')
title('Incoming vs Outgoing Traffic at Junction (Matrix Model)')
legend('Incoming','Outgoing (Linear)')
grid oN
```



Command Window

```
Incoming Traffic (veh/min):
   250
   220
   200
   240

Column sums of T (should be 1):
     1     1     1     1
```

CONGESTION EFFECT (NONLINEAR MODEL)

CONGESTION MODELLING

In real traffic systems, roads have **limited capacity**.

As the number of vehicles increases, traffic flow reduces due to congestion.

To model this behavior, a **nonlinear function** is introduced:

$$F = X\left(1 - \frac{X}{C}\right)$$

Where:

- **X** = traffic flow from the linear model
- **C** = road capacity
- **F** = traffic flow considering congestion

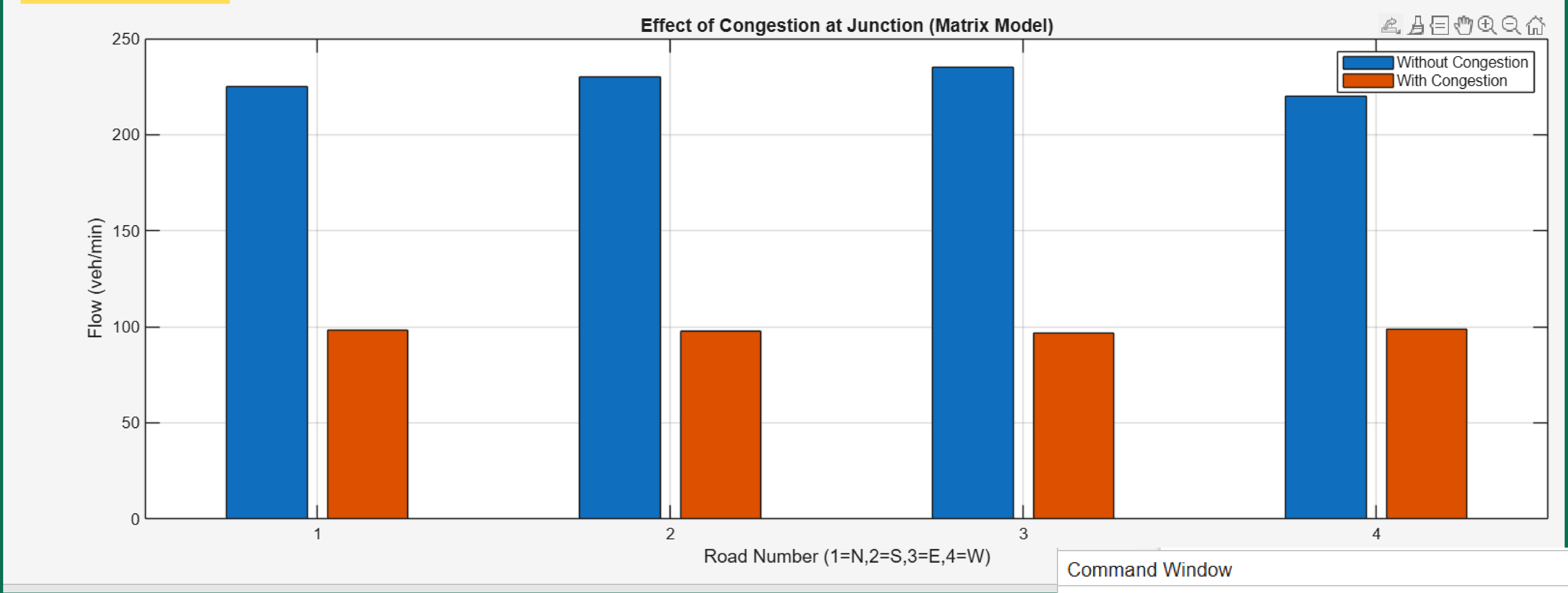
Key Idea

- When traffic is **low**, flow remains almost unchanged.
- As traffic approaches **road capacity**, congestion reduces the effective flow.

CODE

```
%% ---------------------------------
% PART 2 : Congestion Effect (Nonlinear)  (LAB 5 concept)
%% ---------------------------------

capacity = 400;  % congestion parameter  - vehicle/min

% Nonlinear congestion effect
X_out_cong = X_out_linear .* (1 - X_out_linear/capacity);

disp('Outgoing Traffic with Congestion (veh/min):')
disp(X_out_cong)

% Plot comparison
figure
bar([X_out_linear X_out_cong])
legend('Without Congestion','With Congestion')
xlabel('Road Number (1=N,2=S,3=E,4=W)')
ylabel('Flow (veh/min)')
title('Effect of Congestion at Junction (Matrix Model)')
grid on
```

OUTPUT



Command Window

```
Outgoing Traffic without Congestion (veh/min):
   225
   230
   235
   220

Outgoing Traffic with Congestion (veh/min):
   98.4375
   97.7500
   96.9375
   99.0000
```

KINETIC THEORY MODEL

Kinetic Theory Model

Traffic flow can be analyzed using concepts similar to **kinetic theory in physics**, where vehicles behave like moving particles.

The relationship between **velocity and density** is given by:

$$v = v_{max}\left(1 - \frac{k}{k_{max}}\right)$$

Where:

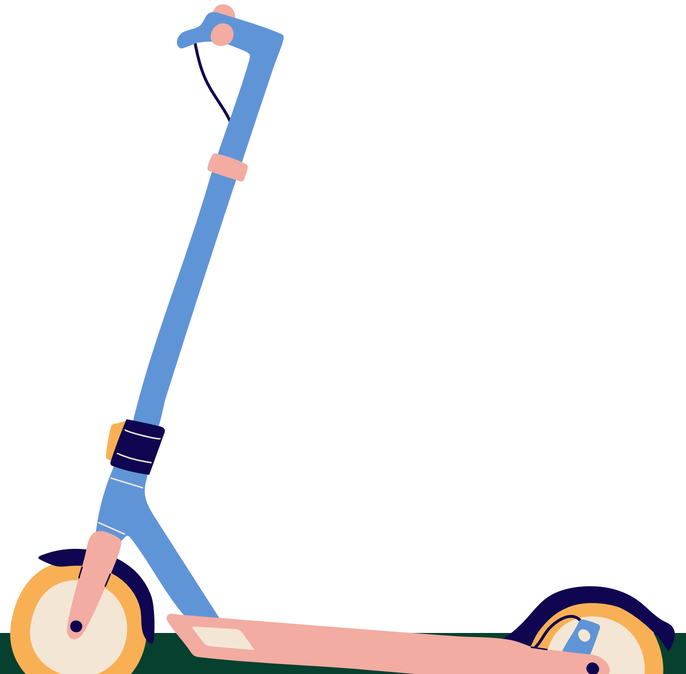
- **v** = vehicle velocity
- **vmax** = maximum velocity
- **k** = traffic density (vehicles/km)
- **kmax** = maximum density

Traffic flow is defined as:

$$q = k \times v$$

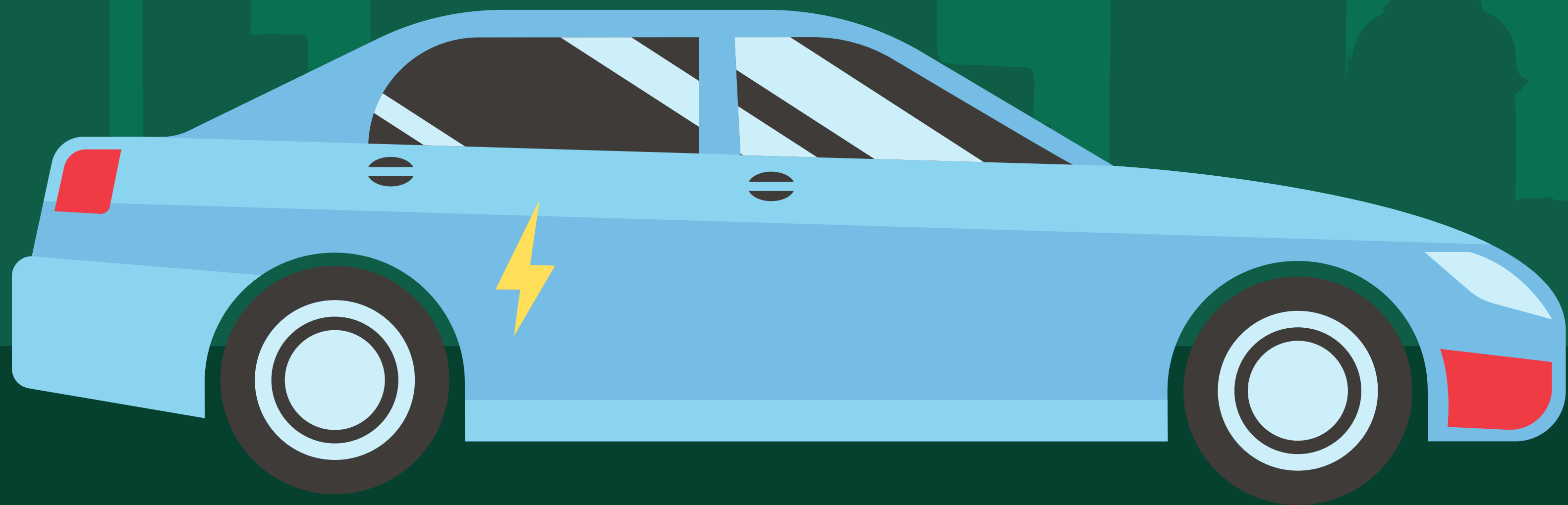
Where:

- **q** = traffic flow (vehicles/hour)



Key Observation

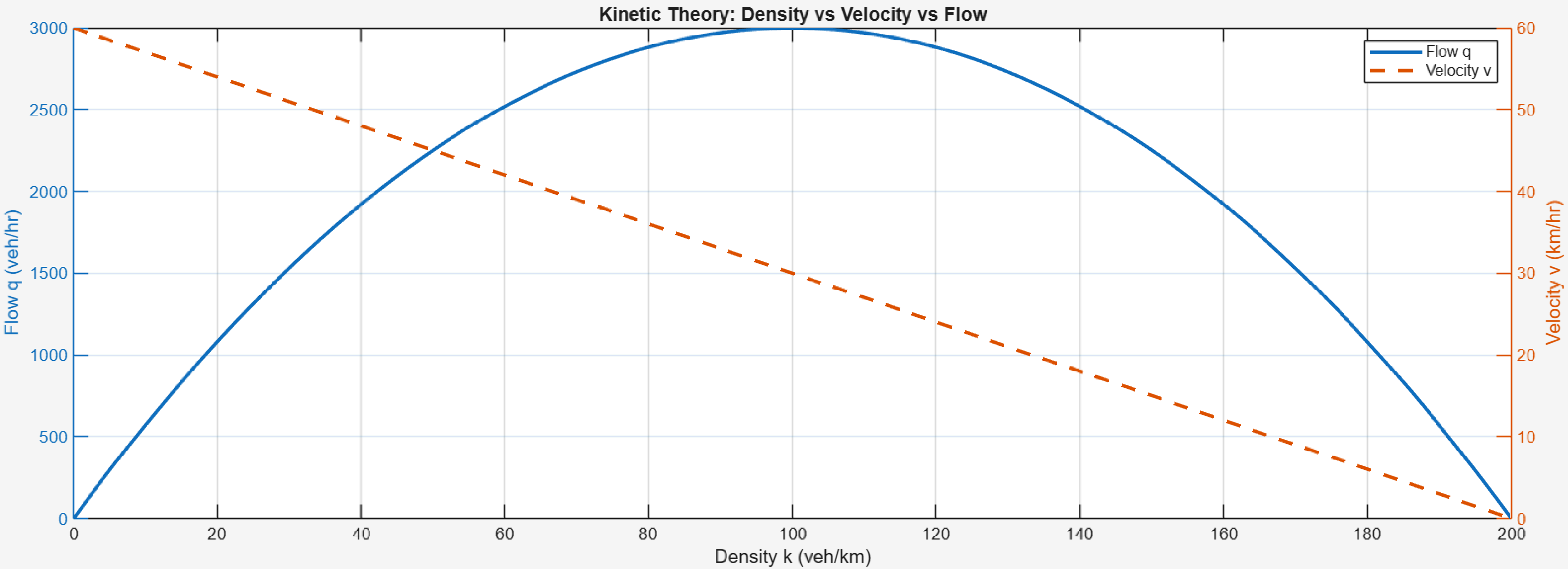
- At **low density,** vehicles move fast but flow is small.
- At **moderate density**, traffic flow reaches a **maximum.**
- At **high density**, congestion reduces both velocity and flow.



CODE

```matlab
%% ------------------------------
% PART 3 : Kinetic Theory Model
%% ------------------------------

k = linspace(0,200,100);  % Density
vmax = 60;
kmax = 200;

v = vmax*(1 - k/kmax);     % Velocity
q = k.*v;                  % Flow

figure
yyaxis left
plot(k,q,'LineWidth',2)
ylabel('Flow q (veh/hr)')

yyaxis right
plot(k,v,'--','LineWidth',2)
ylabel('Velocity v (km/hr)')

xlabel('Density k (veh/km)')
title('Kinetic Theory: Density vs Velocity vs Flow')
legend('Flow q','Velocity v')
grid on
```

OUTPUT


Kinetic Theory: Density vs Velocity vs Flow
Flow q
Velocity v
Flow q (veh/hr)
Velocity v (km/hr)
Density k (veh/km)
3000
2500
2000
1500
1000
500
0
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
180
200

CONCLUSION

- Traffic flow at a four-way junction was **modeled using the conservation of vehicles principle.**
- A **matrix formulation was used to compute outgoing traffic** from incoming traffic.
- A **nonlinear congestion model was introduced** to represent road capacity limitations.
- The **kinetic theory model** explained the relationship between **traffic density, velocity, and flow.**
- **MATLAB simulations and graphs** helped visualize and analyze the behavior of traffic at the junction.


THANK YOU